Introduction

This thesis studies randomized numerical linear algebra and mixed-precision arithmetic with emphasis on covariance-based dependence correction. The motivation is simple. Many linear algebra problems arising in scientific computing are large enough that classical dense methods become expensive, and modern hardware no longer offers one uniform arithmetic. Instead, several precisions are available, often with very different speed, memory cost, and numerical reliability.

Randomized numerical linear algebra addresses the first difficulty. The basic idea is to replace expensive deterministic operations by randomized approximations that preserve the important geometry of the problem. Typical examples are sketching, random projections, and structured random transforms. These methods are especially natural for overdetermined least-squares problems, where the cost is dominated by the size of the data matrix rather than by the dimension of the solution space.

Mixed-precision arithmetic addresses the second difficulty. Instead of performing the whole computation in one format, one tries to use lower precision where it is cheap and higher precision where it is needed for stability or accuracy. This may reduce runtime and memory use, but it also changes the perturbation model of the algorithm. A method that is reliable in one precision may behave quite differently in another.

In this thesis, we are interested in the interaction of these two ideas. Sketching changes the problem by replacing it with a compressed surrogate. Low precision changes the arithmetic by introducing larger rounding errors. Both effects act on the same operator-theoretic problem, and both are filtered through conditioning. Therefore, they should not be studied separately.

The central mathematical contribution pursued here is a mixed-precision randomized low-rank covariance transport map for multivariate dependence correction. Least squares remains essential, but now as the first model problem in which sketching, conditioning, and finite precision can be analyzed in a clean setting. We therefore begin with randomized least-squares solvers based on Gaussian sketches and structured transforms such as the subsampled randomized Hadamard transform, and then transfer the same perturbation viewpoint to regularized covariance operators and the matrix functions used in dependence correction.

The climate-model application motivates the dependence-correction problem, while the thesis focuses on the numerical linear algebra needed to make such a correction analyzable and computationally feasible. Its role is to test whether methods developed on synthetic and theoretical examples remain useful in a more realistic workflow. The outcome need not be a pure speedup. Reduced memory use, acceptable lower-precision behavior, or a clearer picture of failure regimes are also valuable results.

Main Contribution

The thesis advances three interlocking contributions.

1. **Perturbation analysis for regularized covariance transport.** We give deterministic first-order bounds for the perturbation of the regularized transport map $T_\lambda = (C_X + \lambda I)^{-1/2}(C_Y + \lambda I)^{1/2}$ under additive disturbances to the source and target covariance operators. The bounds make explicit how the inverse-square-root amplification factor $K_\lambda(C_X, C_Y)$ depends on the regularized spectral gap, and they apply whether the perturbations arise from randomized low-rank compression, floating-point arithmetic, or both.
2. **Randomized low-rank approximation and mixed-precision error decomposition.** We decompose the total covariance perturbation into a randomized truncation term governed by the spectral tail beyond the chosen rank, and a floating-point term governed by the unit roundoff, the accumulation length, and the norm of the sketching operator. This decomposition is the natural quantitative setting for the balancing-inexactness principle: mixed precision is safe only while $\|E_{\mathrm{fp}}\| \lesssim \|E_{\mathrm{rand}}\|$, and the regime boundary is computable from the rank, the oversampling, and the spectrum of the covariance operator.
3. **Synthetic and climate-shaped experiments identifying rank, regularization, and precision regimes.** We design experiments that vary the approximation rank, the regularization level λ, and the working precision in the sketch products, while tracking covariance error, transport-map error, and the induced dependence error on corrected future data. The experiments distinguish three regimes: an approximation-dominated regime in which lower precision is hidden beneath truncation error, a regularization-dominated regime in which transport instability is the bottleneck, and a precision-visible regime in which floating-point error overtakes the randomized approximation. Negative or inconclusive randomized results are framed as controlled regime-identification findings rather than as failed speedup attempts.

Taken together, these contributions turn the experimental observation that randomized covariance transport is numerically viable only in certain parameter regimes into a coherent thesis-level claim: the regime boundary is predictable from the spectral structure of the covariance operators and the regularization parameter, and mixed precision is safe only inside the approximation-dominated sub-regime.

The structure of the thesis is as follows. The preliminary chapters fix notation and collect the basic mathematical material. The next chapters discuss mixed precision and randomized least squares in more detail, first as model problems and then as technical tools. They are followed by a chapter on randomized covariance transport, where the main dependence-correction map is formulated and its perturbation structure is analyzed. The numerical experiments then test both the least-squares and covariance-transport viewpoints. The final substantive chapter returns to the climate-model application.